Several days later your crew transmit back to Earth a detailed description of your Hades explorations, including motion pictures of you being stretched by the tidal force and pictures of atoms falling into the hole. The description will require 26 years to cover the 26 light-year distance to Earth, and when it finally arrives it will be published with great fanfare by the World Geographic Society.

In their transmission the crew describe your plan for a voyage to the center of the Milky Way: Your starship's rocket engines will blast all the way with a l-g acceleration, so that you and your crew can experience a comfortable 1-Earth-gravity force inside the starship. The ship will accelerate toward the galactic center for half the journey, then it will rotate 180 degrees and decelerate at 1 *g* for the second half. The entire trip of 30,100 light-years distance will require 30,102 years as measured on Earth; but as measured on the starship it will require only 20 years. In accordance with Einstein's laws of special relativity,[26] your ship's high speed will cause time, as measured on the ship, to "dilate"; and this *time dilation* (or *time warp),* in effect, will make the starship behave like a time machine, projecting you far into the Earth's future while you age only a modest amount.[27]

You explain to the World Geographic Society that your next transmission will come from the vicinity of the galactic center, after you have explored its million-solar-mass hole, Sagittario. Members of the Society must go into deep-freeze hibernation for 60,186 years if they wish to live to receive your transmission (30,102 –26 = 30,076 years from the time they receive your message until you reach the galactic center, plus 30,110 years while your next transmission travels from the galactic center to Earth).

Sagittario

After a 20-year voyage as measured in starship time, your ship decelerates into the Milky Way's center. There in the distance you see a rich mixture of gas and dust flowing inward from all directions toward an enormous black hole. Kares adjusts the rocket blast to bring the starship into a coasting, circular orbit well above the horizon. By measuring the circumference and